

Multi-Sensor Fusion for Wheat Above-Ground Biomass Estimation: Sensor Contribution Analysis Under Leave-One-Site-Out Spatial Validation

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Abstract—Accurate estimation of wheat above-ground biomass (AGB) from remote sensing is essential for precision agriculture and yield forecasting. Although multi-sensor fusion of synthetic aperture radar (SAR) and optical imagery is widely adopted, most studies rely on random cross-validation, which inflates generalization metrics due to spatial autocorrelation. Here, we present a spatially rigorous leave-one-site-out (LOSO) evaluation of AGB estimation from Sentinel-1 (S1), Sentinel-2 (S2), PlanetScope (PS), and climate-derived predictors across four wheat site-seasons in Mediterranean Chile (153 destructive samples; AGB range 0–43.5 t/ha). A sensor ablation study with three machine learning algorithms—Random Forest, XGBoost, and GLMnet—reveals that optical S2 predictors provide the dominant improvement over a thermal-time baseline (23.4% RMSE reduction), while S1 SAR contributes negligibly (−0.5%) and PlanetScope adds a modest 5.7%. The best configuration—full-sensor fusion with Random Forest—achieves $\text{RMSE} = 7.10 \pm 2.57 \text{ t/ha}$ and $R^2 = 0.729$ under LOSO. A $\text{GDD} \times \text{canopy-greenness}$ interaction analysis further demonstrates that cumulative thermal time alone underestimates high-biomass crops, justifying multi-source optical fusion. These findings challenge the assumption that SAR systematically improves optical AGB estimation in rain-limited Mediterranean environments.

Index Terms—above-ground biomass, wheat, Sentinel-1, Sentinel-2, PlanetScope, sensor fusion, leave-one-site-out, spatial cross-validation, ablation study, machine learning

I. INTRODUCTION

WHEAT (*Triticum aestivum* L.) contributes approximately 20% of global caloric intake, making timely and accurate crop biomass monitoring a strategic priority for food security [1]. Above-ground biomass (AGB) is a central state variable for crop growth models and yield gap analyses, yet its non-destructive, large-area estimation remains a challenge [2].

Multispectral satellite imagery has long underpinned remote sensing-based AGB estimation, with vegetation indices such as NDVI and EVI serving as proxies for canopy greenness

and leaf area [3]. The availability of free, synoptic Sentinel-2 (S2) data at 10–20 m resolution has substantially advanced this field [4]. Concurrently, Sentinel-1 (S1) SAR (Synthetic Aperture Radar) offers cloud-independent acquisition and sensitivity to canopy structure through its C-band VV/VH backscatter, motivating a growing literature on multi-sensor optical–SAR fusion for biomass estimation [5], [6]. Commercial constellation data such as PlanetScope (PS), with near-daily revisit at 3 m resolution, further extend temporal and spatial coverage [7].

Despite this abundance of sensors, a critical methodological gap persists: most published AGB models are evaluated with random train–test splits that pool observations across sites and seasons [8]. Because field samples collected at the same location across dates share spatial structure, random splitting allows spatially proximate points to appear in both training and test sets, artificially inflating performance metrics and concealing poor generalization to novel sites [9]. Spatial cross-validation—and specifically, leave-one-site-out (LOSO) evaluation—is the appropriate framework for assessing model generalization across agro-ecological contexts, but its application to crop AGB estimation remains rare [10].

A second unresolved question concerns the actual marginal contribution of SAR and very-high-resolution imagery in Mediterranean rain-fed systems. SAR backscatter correlates with canopy water content and structure, but this relationship is confounded by soil moisture and crop architecture at C-band [11], [12]. Studies comparing sensor configurations under the same spatial validation protocol are scarce.

This letter addresses both gaps with three contributions: (i) a spatially rigorous LOSO framework for wheat AGB estimation that prevents inflated metrics from within-site autocorrelation; (ii) a quantitative sensor ablation study isolating the marginal contribution of climate-derived predictors, S2, S1, and PS under identical model conditions; and (iii) agronomy-informed interpretability, including a conditional partial dependence analysis of the interaction between cumulative thermal time (GDD) and canopy greenness.

II. STUDY AREA AND DATA

A. Study Sites and In-Situ Measurements

The study covers three dryland wheat sites in the Mediterranean climatic zone of central Chile (35°–37° S): Hidango (two growing seasons: 2021–22 and 2022–23), La Cancha (2022–23), and Villa Baviera (2020–21), comprising four

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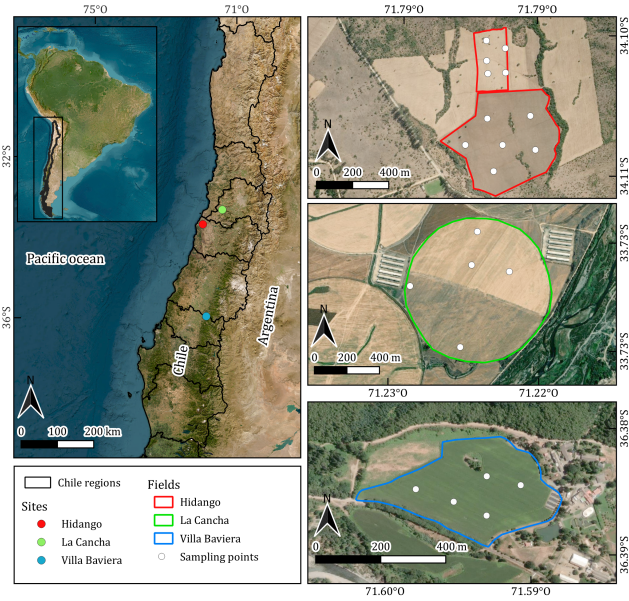


Fig. 1. The map on the left panel shows the location of the sites in Central Chile. The remaining panels illustrate the field boundaries and sampling points for each site. In Hidalgo, the northern field is from the 2021–2022 season, and the southern one from 2022–2023.

distinct site-seasons used as LOSO folds (Fig. 1). Field campaigns collected 153 destructive AGB samples spanning the full phenological cycle from emergence to senescence (AGB range: 0–43.5 t/ha; mean $\pm$ SD: 12.4 $\pm$ 9.7 t/ha). At each sampling date (6–12 per season), three 0.5 m² quadrats per field were harvested, oven-dried at 70 °C for 48 h, and weighed.

B. Satellite Imagery

Sentinel-2 (S2): Level-2A surface reflectance images were retrieved from the Microsoft Planetary Computer via STAC. Twenty-two spectral and vegetation indices were computed per scene, including NDVI, EVI, SAVI, NDWI, LAI, and red-edge chlorophyll indices. Temporal smoothing and cumulative sums were derived to capture seasonal canopy development.

Sentinel-1 (S1): Dual-polarization GRD products (VV, VH) were processed to σ^0 backscatter in linear scale. The cross-polarization ratio VH/VV was also computed as a structural discriminator sensitive to canopy closure and senescence [13].

PlanetScope (PS): Daily 4-band surface reflectance imagery at 3 m resolution was used to compute 15 spectral indices analogous to the S2 set. Indices were smoothed using a Savitzky–Golay filter to reduce cloud contamination.

C. Climate and Soil Moisture

Daily temperature and precipitation data from Chilean AgroMet stations were used to derive: cumulative growing degree days (GDD, base temperature 0 °C), cumulative precipitation, and daily soil moisture estimates from station-level water balance models. GDD is the primary phenological driver of biomass accumulation in wheat [14].

TABLE I
SENSOR ABLATION RECIPES USED IN THE LOSO STUDY

Recipe	Predictors	Purpose
Clima	GDD, precip., soil moist.	Climate-only baseline
S2+Clima	Clima + 22 S2 indices	Optical baseline
S2+S1+Clima	S2+Clima + VV, VH, VH/VV	SAR contribution
Fusion	S2+S1+Clima + 15 PS indices	Full fusion

D. Multicollinearity Audit

A variance inflation factor (VIF) analysis on the full 43-predictor set revealed extreme collinearity in several spectral indices (VIF > 100; e.g., S2_MCARI: 322, PS_B3: 262, GDD cumsum: 142), warranting correlation filtering and regularization as integral preprocessing steps.

III. METHODOLOGY

A. Preprocessing Pipeline

All preprocessing steps were estimated exclusively on training folds to prevent data leakage. The pipeline, implemented in R with `tidymodels` [15], applied: (1) k -NN imputation for missing values due to cloud cover; (2) z -score normalization; and (3) a Pearson correlation filter removing predictors with $|r| \geq 0.9$.

B. Sensor Ablation Recipes

Four preprocessing recipes defined mutually exclusive sensor combinations (Table I). The design allows direct attribution of performance changes to individual sensor layers under otherwise identical conditions.

C. Machine Learning Models

Three algorithms were evaluated with domain-expert fixed hyperparameters to avoid hyperparameter leakage across LOSO folds: Random Forest (RF): trees = 1000, mtry = 7, min_n = 5; XGBoost: trees = 500, max_depth = 4, learning rate = 0.05, subsample = 0.8; GLMnet: penalty = 0.01, mixture = 0.5 (elastic net). This design yields 12 workflows (4 recipes $\times$ 3 models).

D. Leave-One-Site-Out Validation

Spatial generalization was assessed via group-based cross-validation using `group_vfold_cv`, where each fold omits one complete site-season (4 folds total). No observations from a held-out site contribute to training—including preprocessing estimation—eliminating autocorrelation leakage. Reported metrics (RMSE, R², MAE) are means $\pm$ SD across folds. The marginal sensor contribution of PlanetScope is defined as:

$$\Delta_{PS} = \frac{RMSE_{S2+S1+Clima} - RMSE_{Fusion}}{RMSE_{S2+S1+Clima}} \times 100\%, \quad (1)$$

with analogous formulas for S2 and S1 contributions.

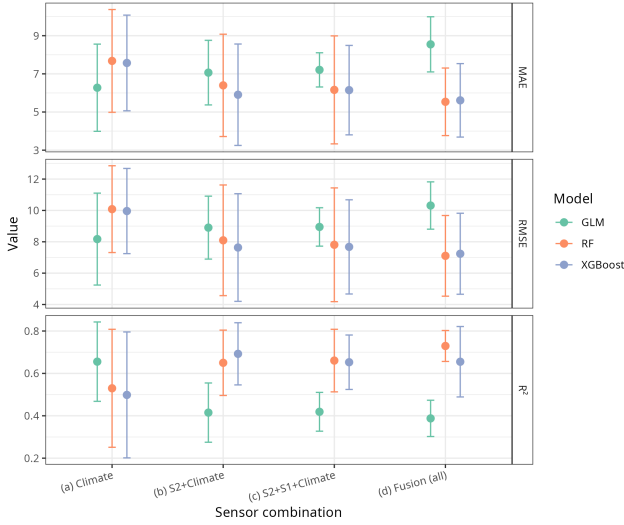


Fig. 2. LOSO ablation metrics (RMSE and R^2) across sensor recipes and algorithms. Dot-and-whisker plots show mean $\pm$ SD over the four site-season folds.

TABLE II
LOS0 PERFORMANCE SUMMARY (MEAN $\pm$ SD, 4 FOLDS). BEST RESULT PER METRIC IN BOLD.

Recipe	Model	RMSE (t/ha)	R^2	MAE (t/ha)
Clima	XGBoost	7.63 $\pm$ 3.43	0.692 $\pm$ 0.147	5.91 $\pm$ 2.66
S2+Clima	RF	8.41 $\pm$ 2.89	0.601 $\pm$ 0.183	6.74 $\pm$ 2.22
S2+Clima	XGBoost	9.96 $\pm$ 2.72	0.499 $\pm$ 0.297	7.57 $\pm$ 2.51
S2+S1+Clima	XGBoost	7.68 $\pm$ 3.00	0.653 $\pm$ 0.128	6.15 $\pm$ 2.34
Fusion	XGBoost	7.24 $\pm$ 2.91	0.714 $\pm$ 0.089	5.68 $\pm$ 2.01
Fusion	RF	7.10 $\pm$ 2.57	0.729 $\pm$ 0.073	5.53 $\pm$ 1.77

E. Model Interpretability

Partial dependence profiles (PDPs) were computed for the four most important predictors (permutation importance on the ensemble model). Agronomically meaningful thresholds—heading (≈ 800 GDD), maturity (≈ 1400 GDD), and canopy closure ($S1_VH = 0.05$)—were annotated on each profile. A conditional PDP assessed the $GDD \times PS_EVI$ interaction by stratifying observations into PS_EVI terciles before computing separate PDPs.

IV. RESULTS AND DISCUSSION

A. LOS0 Ablation Results

Table II reports LOS0 performance. The best configuration is Fusion-RF (RMSE = 7.10 $\pm$ 2.57 t/ha, R^2 = 0.729 $\pm$ 0.073). Notably, the climate-only baseline achieves RMSE = 7.63 $\pm$ 3.43 t/ha for XGBoost, confirming that GDD cumsum alone captures substantial phenological–biomass covariation. Performance differences between recipes are larger for XGBoost, while RF shows more consistent generalization across folds (Fig. 2).

B. Sensor Marginal Contributions

Applying Eq. (1) to the XGBoost results: S2 improves over the climate baseline by 23.4% (RMSE: 9.96 $\rightarrow$ 7.63 t/ha).

Sentinel-1 SAR, however, shows no aggregate benefit—adding VV/VH to S2+Clima marginally worsens XGBoost performance (RMSE: 7.63 $\rightarrow$ 7.68 t/ha, Δ = -0.5%). PlanetScope adds a modest 5.7% improvement over S2+S1 (7.68 $\rightarrow$ 7.24 t/ha).

The negligible SAR contribution is physically interpretable. In the semi-arid Mediterranean climate of the study, crop water stress is frequent after stem elongation, reducing C-band backscatter sensitivity to green biomass and increasing confounding from dry soil backgrounds [11]. This contrasts with wetter temperate systems where S1 consistently aids crop AGB estimation [5], [6]. RF achieves lower fold-to-fold variance than XGBoost (RMSE SD: 2.57 vs. 2.91 t/ha for Fusion), suggesting more robust spatial generalization via bagging.

C. Per-Site Generalization: La Cancha Domain Shift

Fold-level analysis reveals systematic underperformance at La Cancha (RMSE = 8.15 t/ha, bias = -5.26 t/ha), while Hidango and Villa Baviera achieve RMSE ≤ 7.1 t/ha with near-zero bias. La Cancha contributes only one season to the training pool, providing insufficient diversity for the models to learn its edaphic and variety-specific response. All three algorithms overestimate La Cancha biomass, evidenced by empirical prediction interval coverage of 51.4% at that site versus 70.5% at Hidango. This domain shift underscores the need for multi-season data per site when constructing LOS0 folds.

D. GDD $\times$ Canopy Greenness Interaction

The conditional PDP reveals that the GDD–AGB relationship depends strongly on canopy state (Fig. 3). At heading (≈ 800 GDD), the predicted AGB gap between high- PS_EVI (top tercile) and low- PS_EVI (bottom tercile) canopies reaches a mean of 9.41 t/ha (maximum: 26.56 t/ha). Beyond maturity (≈ 1400 GDD), the three terciles converge as senescence reduces optical contrast. This interaction explains why the climate-only recipe underestimates peak-season biomass: GDD accumulation proceeds similarly across fields regardless of yield potential, but canopy density modulates radiation-use efficiency [16]. Multi-source optical fusion captures this interaction; SAR backscatter alone does not.

E. Temporal Bias Structure

Residual analysis as a function of days after sowing (DAS) for Fusion-XGBoost shows a negative LOESS trend from emergence to ≈ 90 DAS (slight underestimation), followed by overestimation during stem elongation (90–150 DAS). Early-season canopies with low fractional cover produce inconsistent vegetation index values depending on soil brightness, while rapid stem elongation creates non-stationary predictor–biomass relationships that models trained on other seasons partially miss.

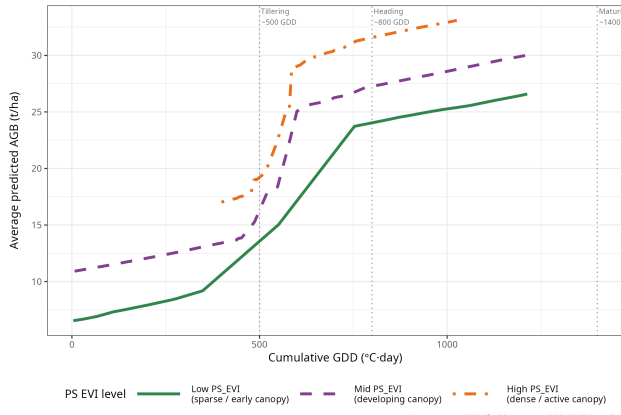


Fig. 3. Conditional partial dependence of predicted AGB on cumulative GDD, stratified by PS_EVI tercile (Low, Mid, High). Vertical dashed lines mark agronomic thresholds: heading (≈ 800 GDD) and maturity (≈ 1400 GDD). The gap between High and Low terciles peaks at heading (mean: 9.41 t/ha; max: 26.56 t/ha), demonstrating that canopy density modulates the thermal-time–biomass relationship.

F. Uncertainty Quantification

Empirical prediction intervals (mean $\pm 1.96 \times \text{SD}$ across RF, XGBoost, and GLMnet Fusion models) achieve 68% global coverage (70.5% Hidango, 51.4% La Cancha; mean width: 18.3 t/ha), well below the nominal 95%. These intervals reflect epistemic uncertainty from algorithm disagreement rather than calibrated distributional predictions; under-coverage is therefore expected, particularly where all algorithms share a systematic bias, as at La Cancha. Conformal or Bayesian calibration methods represent a natural extension [17].

V. CONCLUSION

This letter presents a spatially rigorous, sensor-resolved evaluation of wheat AGB estimation from multi-source remote sensing using LOSO cross-validation across four site-seasons in Mediterranean Chile. The LOSO protocol prevents metric inflation from within-site autocorrelation leakage, which is a critical distinction when reporting model transferability.

The sensor ablation study yields three actionable findings. First, cumulative GDD alone explains substantial biomass variance ($R^2 \approx 0.69$), confirming that phenological timing is the primary AGB driver in Mediterranean wheat systems. Second, Sentinel-2 optical indices provide a 23.4% RMSE improvement over the thermal-time baseline, justifying continued investment in optical monitoring. Third, Sentinel-1 SAR adds negligible predictive value (-0.5% to $+5.7\%$ depending on algorithm) under the semi-arid conditions studied—a finding that challenges generic prescriptions for SAR–optical fusion and suggests that cloud cover frequency and canopy water content determine C-band SAR utility. PlanetScope contributes a modest but consistent 5.7% gain, attributable to higher spatial resolution resolving sub-field variability undetectable at 10 m.

The $\text{GDD} \times \text{PS_EVI}$ conditional PDP demonstrates canopy-state-dependent radiation-use efficiency, producing up

to 9.4 t/ha biomass differences at heading, physically grounding the multi-source fusion strategy.

Limitations and future work. The four available site-seasons limit fold diversity; expanding to 8–10 geographically distinct site-seasons is recommended. L-band SAR data (NISAR, ALOS-4) may improve canopy penetration over C-band in semi-arid conditions. Conformal prediction frameworks should replace the inter-model spread heuristic to provide valid coverage guarantees.

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REFERENCES

- [1] FAO, *The State of Food and Agriculture 2023*. Rome, Italy: FAO, 2023.
- [2] C. Atzberger, “Advances in remote sensing of agriculture: Context description, existing operational monitoring systems and major information needs,” *Remote Sens.*, vol. 5, no. 2, pp. 949–981, 2013. DOI: 10.3390/rs5020949.
- [3] P. J. Zarco-Tejada, A. Miller, G. Morales, A. Berjón, and J. Agüera, “Hyperspectral indices and model simulation for chlorophyll estimation in open-canopy tree crops,” *Remote Sens. Environ.*, vol. 90, no. 4, pp. 463–476, 2004. DOI: 10.1016/j.rse.2003.12.016.
- [4] M. Drusch et al., “Sentinel-2: ESA’s optical high-resolution mission for GMES operational services,” *Remote Sens. Environ.*, vol. 120, pp. 25–36, 2012. DOI: 10.1016/j.rse.2011.11.026.
- [5] G. Macelloni, S. Paloscia, P. Pampaloni, F. Marliani, and M. Gai, “The relationship between the backscattering coefficient and the biomass of narrow and broad leaf crops,” *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 4, pp. 873–884, Apr. 2001. DOI: 10.1109/36.917914.
- [6] C. Xu, Y. Ding, X. Zheng, Y. Qu, Z. Tao, H. Li, and Q. Xie, “Enhanced estimation of crop biomass and height using Sentinel-1 polarization texture indices and integration with optical remote sensing,” *Int. J. Digit. Earth*, vol. 19, no. 1, 2026, Art. no. 2616988. doi: 10.1080/17538947.2026.2616988.
- [7] R. Houborg and M. F. McCabe, “A CubeSat enabled spatio-temporal enhancement method (CESTEM) utilizing Planet, Landsat and MODIS data,” *Remote Sens. Environ.*, vol. 209, pp. 211–226, 2018. doi: 10.1016/j.rse.2018.02.067.
- [8] C. Kumar, G. Walton, P. Santi, and C. Luza, “Random cross-validation produces biased assessment of machine learning performance in regional landslide susceptibility prediction,” *Remote Sens.*, vol. 17, no. 2, Art. no. 213, 2025. doi: 10.3390/rs17020213.
- [9] H. Meyer, C. Reudenbach, T. Hengl, M. Katurji, and T. Nauss, “Improving performance of spatio-temporal machine learning models using forward feature selection and target-oriented validation,” *Environ. Model. Softw.*, vol. 101, pp. 1–9, 2018. DOI: 10.1016/j.envsoft.2017.12.001.
- [10] L. A. Yates, Z. Aandahl, S. A. Richards, and B. W. Brook, “Cross validation for model selection: A review with examples from ecology,” *Ecol. Monogr.*, vol. 93, no. 1, 2023, Art. no. e1557. doi: 10.1002/ecm.1557.
- [11] H. McNairn and B. Brisco, “The application of C-band polarimetric SAR for agriculture: A review,” *Can. J. Remote Sens.*, vol. 30, no. 5, pp. 525–542, 2004. DOI: 10.5589/m04-013.
- [12] E. Ghafari, J. P. Walker, N. N. Das, K. Davary, A. Faridhosseini, X. Wu, and L. Zhu, “On the impact of C-band in place of L-band radar for SMAP downscaling,” *Remote Sens. Environ.*, vol. 251, Art. no. 112111, 2020. doi: 10.1016/j.rse.2020.112111.
- [13] A. Veloso, S. Mermoz, A. Bouvet, T. Le Toan, M. Planells, J.-F. Dejoux, and E. Ceschia, “Understanding the temporal behavior of crops using Sentinel-1 and Sentinel-2-like data for agricultural applications,” *Remote Sens. Environ.*, vol. 199, pp. 415–426, 2017. doi: 10.1016/j.rse.2017.07.015.

- [14] P. D. Jamieson, J. R. Porter, J. Goudriaan, J. T. Ritchie, H. van Keulen, and W. Stol, "A comparison of the models AFRCWHEAT2, CERES-Wheat, Sirius, SUCROS2 and SWHEAT with measurements from wheat grown under drought," *Field Crops Res.*, vol. 55, pp. 23–44, 1998. DOI: 10.1016/S0378-4290(97)00060-9.
- [15] M. Kuhn and J. Silge, *Tidy Modeling with R*. Sebastopol, CA, USA: O'Reilly Media, 2022. [Online]. Available: <https://www.tmw.org>
- [16] J. L. Monteith, "Solar radiation and productivity in tropical ecosystems," *J. Appl. Ecol.*, vol. 9, no. 3, pp. 747–766, 1972. DOI: 10.2307/2401901.
- [17] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. New York, NY, USA: Springer, 2005. DOI: 10.1007/b106715.